



POSTULATE



► All other details of the reservoir are irrelevant. The system knows nothing about the reservoir except this single number ϵ

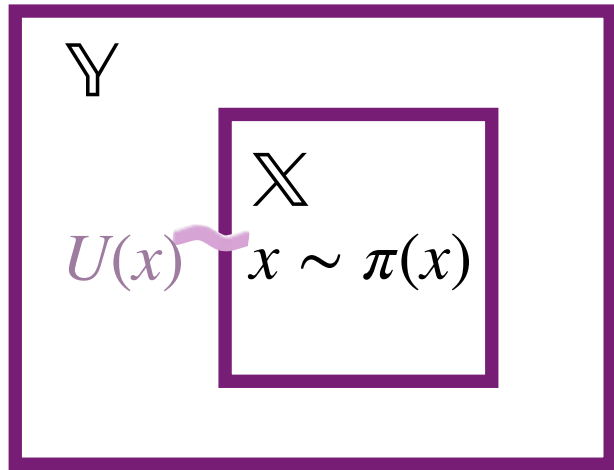
- **Postulate 3:** Given a reservoir state y , we observe a single summary of its effect on the system, the mean energy $\bar{U}$ depending on it induces.

$$\pi[U] = u$$

The system and reservoir are dynamically coupled and will fluctuate even at stationarity

$$\begin{array}{c} X \\ x \sim \eta(x) \end{array}$$

$$\xrightarrow[\mathbb{E}[U] = u]{y \in Y}$$



▶ We can only measure the expected change in energy added or removed from the system

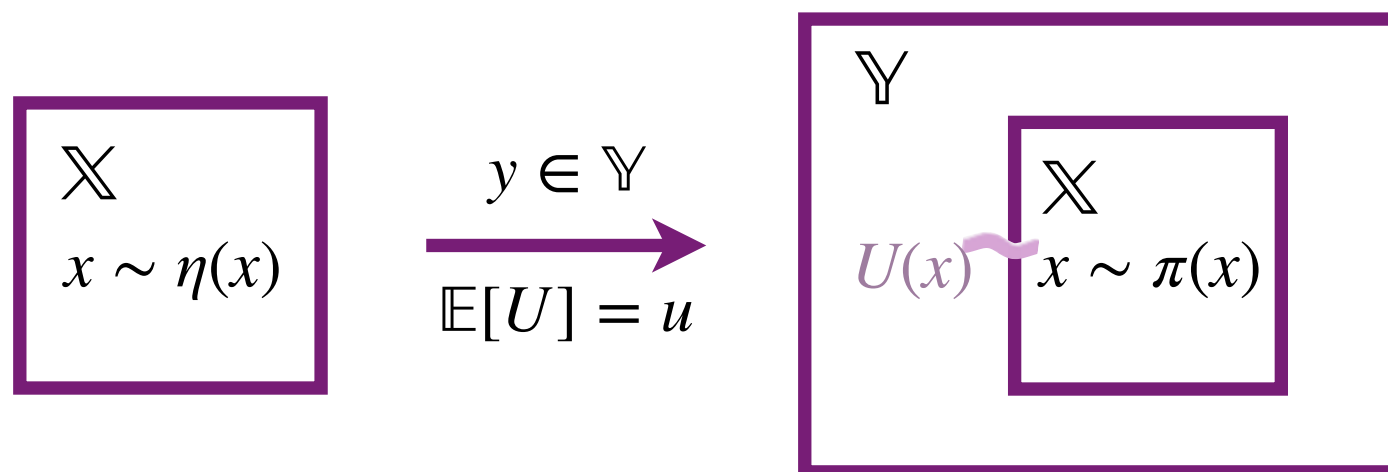
POSTULATE 3

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- ▶ The system and reservoir state dynamically evolve and will fluctuate even at stationarity
- ▶ We can only measure the expected change in energy added or removed from the system
- ▶ **Postulate 3:** Given a reservoir state y , we observe a single summary of its effect on the system, the mean energy $\bar{U}$ depending on it induces.

$$\pi[U] = u$$

- ▶ All other details of the reservoir are irrelevant. The system knows nothing about the reservoir except this single number u



POSTULATE 4

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